CSE 270 Sports Analytics

Predictions based on Performance Ratings

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Predictions based on Performance Ratings

Libraries used

```
library(dplyr)
library(elo)
library(ggplot2)
library(SportsAnalytics270)
library(caret)
```

Predictions based on Performance Ratings

Elo ratings

Elo rating system

- Is initially developed by Hungarian-american professor of physics Arpad Elo.
- Initially was used to rank chess players.
- Can be used to predict the game outcome for any two-players or two-teams game.



Arpad Elo

Elo rating system

- The basic idea
- Assign performance ratings to each team/player. This rating is going to show the ability of the team to win.
- After each game update the the player's rating based on the outcome.
- Use the ability scores to predict the match outcome of the game

Elo rating system

The general formula

The expected probability of winning a game by Team A will be calculated by the following formula.

For home team

$$Y^H = \frac{1}{1+c^{(R^A-\theta-R^H)/d}}$$

For away team

$$Y^A = 1 - \frac{1}{1+c^{(R^A-\theta-R^H)/d}}$$

Where R^H, R^A are home and away teams Ratings, θ is some context parameter (can be omitted) such as home team advantage.
The parameters c and d are used to scale the ratings and are usually taken as c=10, d=400

Elo rating system

- Why are c=10, d=400.
- This is just for convention and scaling. If Player A is 10 times more likely to beat Player B (Odds), then the difference in the Elo scores needs to be 400.
- Assume teams playing in a neutral field $\theta = 0$

$$Y^H = \frac{1}{1+10^{(R^A-R^H)/400}}, Y^A = 1 - \frac{1}{1+10^{(R^A-R^H)/400}}$$

$$Odds(\text{H beats A}) = \frac{Y^H}{Y^A} = \frac{\frac{1}{1+10^{(R^A-R^H)/400}}}{1-\frac{1}{1+10^{(R^A-R^H)/400}}} = \frac{\frac{1}{1+10^{(R^A-R^H)/400}}}{\frac{10^{(R^A-R^H)/400}}{1+10^{(R^A-R^H)/400}}} = 10^{(R^H-R^A)/400}$$

If $R^H - R^A = 400$, then $Odds(\text{H beats A}) = 10$

Elo rating system

- Several R packages for elo rating calculations
- We will use mostly “elo”

```
install.packages("PlayerRatings")
install.packages("elo")
install.packages("EloRating")
```

Elo rating system

- predict probability to win for player A and player B
- Calculate the odds to win

```
elo.prob(elo.A=1500, elo.B=1500)
```

```
## [1] 0.5
```

```
elo.prob(elo.A=1900, elo.B=1500)
```

```
## [1] 0.9090909
```

Elo rating system

c = 10, d = 400 parameters

```
elo.prob(elo.A=1900, elo.B=1500)
```

```
## [1] 0.9090909
```

```
# The odds
elo.prob(elo.A=1900, elo.B=1500)/(1-elo.prob(elo.A=1900, elo.B=1500))
```

```
## [1] 10
```

With the difference of 400 Elo rating points, it is 10 times more likely that Team A will win

Elo rating system

The rating is updated with the following formula

$R_T^H = R_{t-1}^H + K * (\alpha^H - y^H)$

where

R_t^H updated rating for the home team

R_{t-1}^H rating before the game

α^H is the actual outcome for the game

y^H is the predicted probability of Home team to win

K is a positive number

Elo rating system

The actual score for the home team α^H is given by

$$\alpha^H = \begin{cases} 1, \textit{HomeTeamWon} \\ 0.5, \textit{Draw} \\ 0, \textit{Otherwise} \end{cases}$$

In football/soccer the coefficient K can be updated based on the goal differential

$$K = k_0(1+\delta)^\lambda$$

Where λ is the absolute goal difference, k_0, λ are positive constants There are several suggestions for the value of K, we will take it equal to 20.

Elo rating system

example: Two players

- Player A, Elo rating of 1500
- Player B, Elo rating of 1600

$$R_T^H = R_{t-1}^H + K * (\alpha^H - y^H)$$

```
# prob for A
elo.prob(elo.A=1500, elo.B=1600)
```

```
## [1] 0.359935
```

```
1500+20*(1-0.359935)
```

```
## [1] 1512.801
```

```
elo.prob(elo.A=1600, elo.B=1500)
```

```
## [1] 0.640065
```

```
1600+20*(0-0.640065)
```

```
## [1] 1587.199
```

Elo rating system

- The player A will get an increase in rating by 12.8013
- The player B will get a decrease in rating by 12.8013
- 12.8013 is calculated using function elo.update()

```
elo.calc(elo.A=1500, elo.B=1600, wins.A=1, k = 20)
```

```
##      elo.A    elo.B
## 1 1512.801 1587.199
```

```
elo.update(elo.A=1500,elo.B=1600,wins.A=1, k=20)
```

```
## [1] 12.8013
```

Elo rating system

Dataset on EPL games for seasons 2000-2022

```
data("epl_00_22")
str(epl_00_22)
```

```
## 'data.frame':    8360 obs. of  6 variables:
##  $ Home  : chr  "Charlton Athletic" "Chelsea FC" "Coventry City" "Derby County" .
##  $ Away  : chr  "Manchester City" "West Ham United" "Middlesbrough FC" "Southampt
##  $ HG    : num  4 4 1 2 2 0 1 1 3 2 ...
##  $ AG    : num  0 2 3 2 0 0 0 0 1 0 ...
##  $ Round : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Season: num  2001 2001 2001 2001 2001 ...
```

Elo rating system

score function from Elo package does a very simple calculation: Based on the game score returns a vector with the results, 1 if team A won, 0.5 if it was a draw and 0
if Team A lost

3 games, 2-0, 0-0, 0-1

```
score(2,0)
```

```
## [1] 1
```

```
score(0,0)
```

```
## [1] 0.5
```

```
score(0,1)
```

```
## [1] 0
```

Elo rating system

Run Elo model

```
elos <- elo.run(score(HG, AG) ~ Home + Away,
        data = epl_00_22, k = 20)
elos_df <- as.data.frame(elos)
```

score(HG, AG) calculates the final score of the game based on Home Goal and Away Goal statistics

Elo rating system: Choosing K

- K values can be updated based on the importance of the games
- The Eloratings.net uses the following values for k for national teams

Competition	Multiplier
Friendly match	20
All minor tournaments	30
World Cup and continental qualifiers and major tournaments	40
Continental championship finals and major intercontinental tournaments	50
FIFA World Cup finals	60

Elo rating system

- Look at the dataframe
- initial values for Elo are set to 1500

```
head(elos_df)
```

```
##                team.A           team.B p.A wins.A update.A update.B elo.A elo.B
## 1 Charlton Athletic  Manchester City 0.5    1.0       10      -10  1510  1490
## 2        Chelsea FC  West Ham United 0.5    1.0       10      -10  1510  1490
## 3     Coventry City Middlesbrough FC 0.5    0.0      -10       10  1490  1510
## 4      Derby County   Southampton FC 0.5    0.5        0        0  1500  1500
## 5      Leeds United       Everton FC 0.5    1.0       10      -10  1510  1490
## 6    Leicester City      Aston Villa 0.5    0.5        0        0  1500  1500
```

Elo rating system

- The final elo ratings
- n=10 as there are 10 games each round

```
tail(elos_df, n=10)
```

```
##                     team.A                 team.B       p.A wins.A   update.A
## 8351             Arsenal FC             Everton FC 0.6738619      1   6.522761
## 8352           Brentford FC          Leeds United 0.5330456      0 -10.660911
## 8353 Brighton & Hove Albion       West Ham United 0.4469655      1  11.060690
## 8354             Burnley FC      Newcastle United 0.3947975      0  -7.895951
## 8355             Chelsea FC            Watford FC 0.8386514      1   3.226973
## 8356         Crystal Palace     Manchester United 0.3476766      1  13.046469
## 8357         Leicester City        Southampton FC 0.6071852      1   7.856297
## 8358           Liverpool FC Wolverhampton Wanderers 0.8250725      1   3.498550
## 8359        Manchester City           Aston Villa 0.8511795      1   2.976411
## 8360           Norwich City     Tottenham Hotspur 0.1496164      0  -2.992328
##        update.B    elo.A    elo.B
## 8351  -6.522761 1654.759 1515.646
## 8352  10.660911 1523.598 1521.924
## 8353 -11.060690 1570.297 1585.167
## 8354   7.895951 1490.213 1580.216
## 8355  -3.226973 1690.371 1397.591
## 8356 -13.046469 1549.612 1632.834
## 8357  -7.856297 1588.227 1496.861
## 8358  -3.498550 1818.233 1541.782
## 8359  -2.976411 1833.259 1524.363
## 8360   2.992328 1362.177 1670.015
```

Elo rating system

Get the final elos

```
elo_final <- final.elos(elos)
sort(elo_final, decreasing = T)
```

```
##         Manchester City            Liverpool FC             Chelsea FC
##                1833.259                1818.233                1690.371
##       Tottenham Hotspur              Arsenal FC       Manchester United
##                1670.015                1654.759                1632.834
##          Leicester City        West Ham United        Newcastle United
##                1588.227                1585.167                1580.216
##  Brighton & Hove Albion         Crystal Palace Wolverhampton Wanderers
##                1570.297                1549.612                1541.782
##             Aston Villa            Brentford FC            Leeds United
##                1524.363                1523.598                1521.924
##              Everton FC         Southampton FC             Burnley FC
##                1515.646                1496.861                1490.213
##          Wigan Athletic              Stoke City         Birmingham City
##                1488.608                1484.236                1483.599
##            Swansea City            Blackpool FC        Bolton Wanderers
##                1473.755                1471.149                1467.713
##         AFC Bournemouth            Ipswich Town       Charlton Athletic
##                1464.198                1463.073                1458.265
##        Blackburn Rovers              Hull City        Sheffield United
##                1447.924                1441.610                1437.729
##            Cardiff City    West Bromwich Albion           Portsmouth FC
##                1434.696                1433.524                1431.484
##          Coventry City             Reading FC          Sunderland AFC
##                1428.123                1423.624                1418.103
##              Fulham FC       Middlesbrough FC     Queens Park Rangers
##                1416.552                1415.869                1405.656
##             Watford FC          Bradford City       Huddersfield Town
##                1397.591                1397.343                1371.415
##            Norwich City            Derby County
##                1362.177                1294.607
```

Elo rating system

- Taking into account Home advantage
- θ Home advantage parameter $= 200$
- This means that for every game, 200 is added to the home teams Elo rating

$$Y^H = \frac{1}{1+c^{(R^A - 200 - R^H)/d}}$$

$$Y^A = 1 - \frac{1}{1+c^{(R^A - 200 - R^H)/d}}$$

Elo rating system

```
epl2 <- elo.run(score(HG, AG) ~ adjust(Home, 200) +
                                Away,  data = epl_00_22, k = 20)
```

```
sort(final.elos(epl2), decreasing = T)
```

```
##         Manchester City            Liverpool FC             Chelsea FC
##                1909.845                1894.193                1742.336
##       Tottenham Hotspur              Arsenal FC       Manchester United
##                1707.137                1692.563                1683.095
##          Leicester City        West Ham United        Newcastle United
##                1612.100                1606.539                1576.566
##  Brighton & Hove Albion         Crystal Palace Wolverhampton Wanderers
##                1567.152                1554.444                1553.167
##               Everton FC            Leeds United            Brentford FC
##                1527.636                1519.899                1518.926
##             Aston Villa          Southampton FC             Burnley FC
##                1517.872                1500.094                1487.025
##              Stoke City          Wigan Athletic         Birmingham City
##                1484.091                1478.457                1476.486
##            Swansea City            Ipswich Town            Blackpool FC
##                1467.436                1464.047                1460.175
##         AFC Bournemouth        Bolton Wanderers       Charlton Athletic
##                1456.310                1450.821                1443.971
##        Blackburn Rovers       Sheffield United           Portsmouth FC
##                1440.777                1424.196                1421.569
##             Cardiff City           Coventry City               Hull City
##                1419.298                1417.951                1416.978
##    West Bromwich Albion             Reading FC          Sunderland AFC
##                1416.249                1407.307                1401.467
##        Middlesbrough FC              Fulham FC     Queens Park Rangers
##                1398.166                1392.767                1384.800
##           Bradford City             Watford FC       Huddersfield Town
##                1384.045                1377.189                1354.533
##            Norwich City          Derby County
##                1328.660                1261.665
```

Elo rating system

Defining dynamic K factor

Update the Elo rating formula using the following equation

$$R_t^H = R_{t-1}^H + K * G * (\alpha^H - y^H)$$

Where G is a goal difference factor. The website Eloratings.net uses the following formula for G Factor.

$$G = \begin{cases} 1, N <= 1 \\ 1.5, N = 2 \\ (N + 11)/8, N >= 3 \end{cases}$$

Elo rating system

Adding the goal difference factor

```
epl_00_22$G <- ifelse(abs(epl_00_22$HG-epl_00_22$AG)<=1, 1,
                      ifelse(abs(epl_00_22$HG-epl_00_22$AG)==2,1.5,
                             (abs(epl_00_22$HG-epl_00_22$AG)+11)/8))
```

```
head(epl_00_22,5)
```

```
##                 Home              Away HG AG Round Season     G
## 1 Charlton Athletic   Manchester City  4  0     1   2001 1.875
## 2        Chelsea FC   West Ham United  4  2     1   2001 1.500
## 3     Coventry City Middlesbrough FC  1  3     1   2001 1.500
## 4      Derby County    Southampton FC  2  2     1   2001 1.000
## 5      Leeds United        Everton FC  2  0     1   2001 1.500
```

Elo rating system

```
epl3<-elo.run(score(HG, AG) ~ adjust(Home, 200) + Away +
                              k(20*epl_00_22$G),  data = epl_00_22 )
```

```
sort(final.elos(epl3), decreasing = T)
```

```
##         Manchester City            Liverpool FC             Chelsea FC
##                1971.677                1960.491                1783.356
##       Tottenham Hotspur              Arsenal FC       Manchester United
##                1744.307                1724.655                1696.029
##         West Ham United          Leicester City        Newcastle United
##                1640.021                1630.497                1593.126
##          Crystal Palace  Brighton & Hove Albion             Aston Villa
##                1586.190                1578.062                1552.191
## Wolverhampton Wanderers            Brentford FC              Everton FC
##                1544.900                1541.228                1520.015
##            Leeds United          Southampton FC              Burnley FC
##                1515.659                1489.420                1482.875
##          Wigan Athletic              Stoke City            Swansea City
##                1479.286                1476.806                1468.772
##         AFC Bournemouth            Ipswich Town         Birmingham City
##                1455.471                1454.764                1454.399
##           Blackpool FC        Blackburn Rovers        Bolton Wanderers
##                1452.081                1440.332                1436.024
##       Charlton Athletic           Portsmouth FC                Hull City
##                1424.656                1420.757                1410.687
##        Sheffield United             Cardiff City           Coventry City
##                1404.830                1400.593                1396.630
##     Queens Park Rangers          Sunderland AFC    West Bromwich Albion
##                1395.745                1394.454                1392.923
##              Reading FC        Middlesbrough FC              Fulham FC
##                1388.352                1388.068                1363.571
##           Bradford City              Watford FC       Huddersfield Town
##                1353.378                1349.938                1329.647
##           Norwich City            Derby County
##                1292.760                1220.377
```

Elo rating system

- Can we predict the actual outcome and not just the probability ?
- Result for the home team

$$\text{Predicted outcome} = \begin{cases} 1, p >= 0.55 \\ 0.5, 0.45 < p < 0.55 \\ 0, p <= 0,45 \end{cases}$$

```
epl2 <- elo.run(score(HG, AG) ~ adjust(Home, 50) +
                                Away,  data = epl_00_22, k = 20)
epl2 <- as.data.frame(epl2)

epl2$Predicted <- ifelse(epl2$p.A>=0.55, 1,
                         ifelse(epl2$p.A<=0.45,0,0.5))
library(caret)
confusion <- table(epl2$wins.A, epl2$Predicted)
```

```
sum(diag(confusion))/nrow(epl2)
```

```
## [1] 0.5136364
```

Elo rating system

- The teams change every season, players come and go
- The rating of the previous season cannot be carried fully to the new season
- The following formula is suggested by the MARS rating system

New Season Starting Rating = 1500+(Previous Season Final Rating-1500)/2

For newly promoted teams

New Season Starting Rating = 1500+(Average Final Rating of relegated teams-1500)/2)

Elo rating system

To get the Elo rating working better

- Tune home advantage parameter
- Tune new teams initial Elo's parameters
- different K parameters for different games
- Think of what else can influence the game

Elo rating system

Lot of different web-sites provides elo ratings in different sports

- Some mix of elo ratings and optimizations from FiveThirtyEight on NBA teams and games https://projects.fivethirtyeight.com/carmelo/
- Elo ratings from nfl https://projects.fivethirtyeight.com/complete-history-of-the-nfl/
- Football team elos
- European clubs elo ratings http://clubelo.com/
- for national teams http://eloratings.net